

kafka-isotope Library

kafka-isotope provides **end-to-end record tracing for Apache Kafka** using Kafka record headers, a **ProducerInterceptor**, and optional Prometheus metrics.

It works by attaching lightweight tracing artifacts—called *isotopes*—to records as they move through Kafka event pipelines.

An isotope is a small tracing payload carried in Kafka record headers. Like a biochemical isotope used to trace molecules through a metabolic pathway, it enables the journey of a record through an event-driven architecture to be observed and analyzed.

On first produce, **kafka-isotope** stamps a UUIDv7 trace ID and origin metadata onto the record. Each subsequent re-produce appends another hop, allowing the full path of a record through Kafka pipelines to be reconstructed.

Built entirely on Kafka's public extension points—a **ProducerInterceptor** plus record headers—**kafka-isotope** requires:

- No broker changes
- No sidecar agents
- No vendor lock-in

From isotope headers, **kafka-isotope** derives trace data that can be analyzed directly, exported as **Prometheus metrics**, or emitted as **OpenTelemetry (OTel) spans** to support:

- End-to-end latency
- Pipeline topology
- Hop distribution
- Drift detection
- Stuck trace detection
- Per-trace forensic replay

This makes **kafka-isotope** a lightweight but powerful observability layer for Kafka-based event-driven architectures.

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1.0 Modules

Module	Coordinate	Pulls	Use it for
kafka-isotope-core	<code>ai.signalroom:kafka-isotope-core</code>	Jackson, SLF4J (<code>kafka-clients</code> is <code>compileOnly</code>)	Trace propagation — interceptor, headers, consume markers.
kafka-isotope-metrics	<code>ai.signalroom:kafka-isotope-metrics</code>	kafka-isotope-core + Micrometer/Prometheus	Optional <code>/metrics</code> exporter for the stateless reports.
kafka-isotope-otel	<code>ai.signalroom:kafka-isotope-otel</code>	kafka-isotope-core + OTLP protobuf	Optional span writer — one <code>OTel span</code> per hop on a Kafka topic, read by a stock OpenTelemetry (OTel) Collector.

`kafka-isotope-core` has no metrics or tracing dependency. It routes every emission through two sinks that start out as no-ops (`NoOpMetricsSink` and `NoOpSpanSink`). Adding `kafka-isotope-metrics` or `kafka-isotope-otel` to the classpath doesn't change that: the real sink is registered only when your application calls `PrometheusIsotopeMetrics.start(port)` or `KafkaOtlpSpanSink.start(config)`, and stays a no-op if that call fails. Until then, each record costs an "is it enabled?" check and no metric or span work.

2.0 Install using Gradle

```

repositories {
    mavenCentral()
}
dependencies {
    implementation 'ai.signalroom:kafka-isotope-core:0.19.0'
    implementation 'ai.signalroom:kafka-isotope-metrics:0.19.0' // optional
    - only for Prometheus metrics
    implementation 'ai.signalroom:kafka-isotope-otel:0.19.0' // optional
    - only for OTel spans
}

```

3.0 Demo

A runnable reference event pipeline (Confluent Platform on minikube / Confluent Cloud, the seven Flink reports, and a one-command Prometheus + Grafana showcase) lives in the companion repo: [j3-signalroom/confluent-kafka-isotope](https://github.com/signalroom/confluent-kafka-isotope).